TrueFalse Challenge

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Welcome to the TrueFalse Puzzle! The goal of this puzzle is to guess a predetermined positive integer n, or to narrow it down to as few choices as possible. You will be given 10 statements, and you have to figure out which are true and which are false. Based on your answer to each statement, you will be given a hint towards finding n. The harder the question, the more restrictive hints. Note that the correct answer doesn't necessarily correspond to the choice that results in fewer possible choices for n. A calculator probably won't help you with these problems, but will be useful for finding n.

Here is an example statement:

0. Preliminary information

True or False Statement

If the statement is true, this is true	If the statement is false, this is true

Now that you know how this all works, take a go at the problems and good luck!

The solution is on page 5.

1.

Given three sets P, Q, and R, $P \cup Q \cup R = P \cap Q \cap R$ if and only if $P = Q = R = \emptyset$.

$10^6 < n < 10^{12}$

$n < 10^9$

2. Let $f(x), g(x)$ be real-valued functions such that $f(x) = \dfrac{1 - x^{100}}{1 + x^2}$ and $g(x) = \dfrac{1 + x^{100}}{1 - x^2}$.

The maximum of $f(x)$ and the minimum of $g(x)$ are the same value.

The last two digits of n in base 2 are 00.

The last three digits of n in base 3 are 000.

3.

There exists at least one non-degenerate circle tangent to any three distinct lines in a plane.

n, when expressed in base 63, has 4 digits.

n, when expressed in base 36, has 5 digits.

4. Let the choose function be defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

$$\sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$

n is one of the terms of an infinite arithemetic sequence with $a_1 = 1$ and $a_2 = 2024$.

n is one of the terms of an infinite geometric sequence with $a_1 = 1$ and $a_{2024} = 2$.

5.

There exists a real number k such that $\cos\left(\frac{2\pi n}{k}\right)$ never repeats a value in the range $[-1, 1]$ across all positive integers n.

Let H_k be the hint you chose for problem k, and N_k be the hint you didn't choose.

Instead of H_3, use hint N_3.	Instead of H_1, use hint N_1.

6. Let $f(x) = x^2 - 8x + 14$ and $g(a, b, p) = \gcd(a + b, \frac{a^p+b^p}{a+b})$, where p is an odd prime. *(Problem Credit: Patrick Was)*

There exists no x_1, x_2 such that $f(x_1) = f(x_2) = p$ and $g(a_1, b_1, p) \neq g(a_2, b_2, p)$ for all integers a_1, b_1, a_2, b_2 that satisfy $a_1 + b_1 = x_1$ and $a_2 + b_2 = x_2$.

n is divisible by 701.	n is divisible by 107.

7. $\triangle ABC$ is an isosceles right triangle with right angle at B. Let point X_1 be the foot of the altitude from B to AC, and let X_2 be the foot of the altitude from X_1 to BC. Then, for positive integer $n > 2$, let X_n be the foot of the altitude from X_{n-1} to segment $X_{n-2}C$. Consider the incenters O_1 of $\triangle ABX_1$, O_2 of $\triangle BX_1X_2$, and O_n of all triangles $\triangle X_nX_{n+1}X_{n+2}$.

For integers $k > 0$, all incenters O_{2k-1} are collinear, all O_{2k} are collinear, and these two lines intersect at C.

Let $b_k = \prod_{i=1}^{k} \left(1 + \frac{1}{i}\right)^2$.

$n \equiv b_{2023} \pmod 9$	$n \equiv b_{2024} \pmod 9$

8. Let $f^n(x)$ be f iterated n times onto itself, so $f^3(x) = f(f(f(x)))$. Consider the function $f(x) = |||x| - \pi| - \pi|$.

$$\lim_{n\to\infty} f^n(x) = \arccos(\cos x).$$

$\log_5 n > 8(\log_5 2) + 9$	$\log_2 n < 9(\log_2 10) - 1$

9. Let $f(x) = \frac{\lceil x \rceil}{\lfloor x \rfloor}$, where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor functions, respectively. The ceiling function returns the least integer greater than or equal to x, and the floor function returns the greatest integer less than or equal to x for any real input x. For example, $\lceil 2.2 \rceil = 3$ and $\lfloor 1.7 \rfloor = 1$.

The following integral converges:

$$\int_1^\infty \left(f(x) - \frac{1}{f(x)} \right) dx$$

$$n \equiv \sum_{k=1}^{18} 2k \binom{18}{k} \pmod{19}$$

$$n \equiv \sum_{k=1}^{20} k \binom{20}{k} \pmod{19}$$

10. Let $f(x, y) = \frac{1}{xy(x^2 + y^2)}$.

The integral

$$\int_1^\infty \int_1^\infty f(x, y) dx dy$$

converges to a value Z that satisfies $e^{nZ} = m$ for some positive integers n and m.

Let S be the set of possible values of n you have from other clues.

n is the element of S with the minimum sum of digits.

n is the element of S with the maximum sum of digits.